

Problem 1 [30 points] Query-Key-Value Mechanics in Self-Attention

Self-attention is the core building block for the Transformer model, which has kickstarted the amazing progress in seq2seq modeling. In this question, you will be studying the detailed mechanics of the Query-Key-Value interaction in a self-attention block in processing a sequence. Here we will be studying the case where the encoder only takes in 3 inputs, with the variables defined as:

$$a_1 = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \quad a_2 = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \quad a_3 = \begin{bmatrix} -2 \\ 6 \end{bmatrix} \quad W_q = \begin{bmatrix} 1 & 0 \\ 2 & 9 \end{bmatrix} \quad W_k = \begin{bmatrix} 0 & 1 \\ 6 & 0 \end{bmatrix} \quad W_v = \begin{bmatrix} 4 & 3 \\ 9 & 1 \end{bmatrix}$$

where h_i denotes the hidden states at timestep i at a arbitrary self-attention layer. Each q_i, k_i, v_i is determined by $q_i = W_q a_i, k_i = W_k a_i, v_i = W_v a_i$.

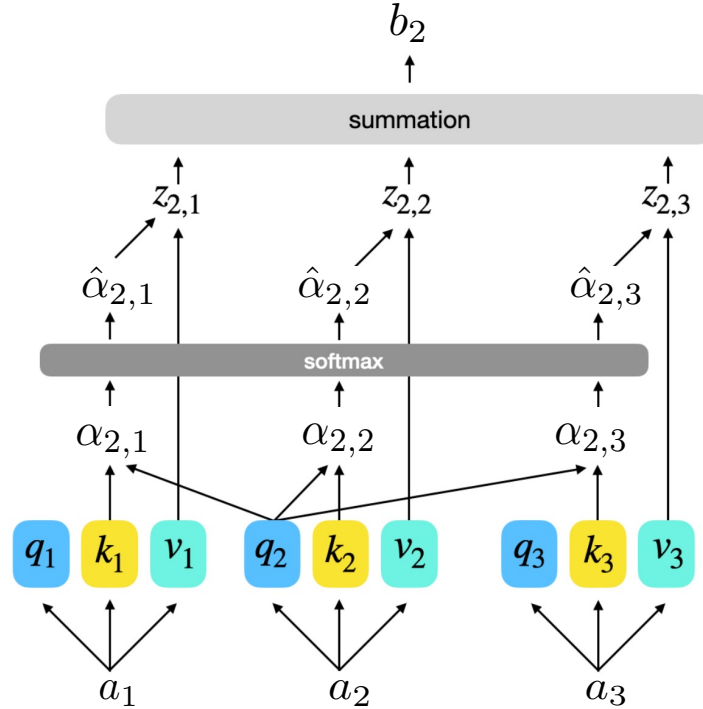


Figure 1: Self-attention of timestep 2 in encoder

1 [10 points]. What are $\alpha_{2,1}, \alpha_{2,2}, \alpha_{2,3}$? Compute their values.

$$\begin{aligned} \alpha_{2,1} &= q_2^T k_1 / \sqrt{d} = (W_q a_2)^T (W_k a_1) / \sqrt{d} \\ &= \begin{pmatrix} 1 & 0 \\ 2 & 9 \end{pmatrix}^T \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{pmatrix} 0 & 1 \\ 6 & 0 \end{pmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} / \sqrt{2} \\ &= \begin{pmatrix} 4 & 35 \end{pmatrix} \begin{bmatrix} 1 \\ 30 \end{bmatrix} / \sqrt{2} = 745.29 \end{aligned} \tag{1}$$

$$\alpha_{2,2} = q_2^T k_2 / \sqrt{d} = (W_q a_2)^T (W_k a_2) / \sqrt{d} = \begin{pmatrix} 4 & 35 \end{pmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} / \sqrt{2} = 602.45 \tag{2}$$

$$\alpha_{2,3}/\sqrt{d} = q_2^T k_3/\sqrt{d} = (W_q a_2)^T (W_k a_3)/\sqrt{d} = \begin{bmatrix} 4 & 35 \end{bmatrix} \begin{bmatrix} 6 \\ -12 \end{bmatrix} / \sqrt{2} = -280.01 \quad (3)$$

2 [10 points]. Figure 1 shows the computational graph of the three inputs in a transformer layer. Write out the operation process from input a_1 , a_2 , a_3 to output b_2 . You don't need to calculate the values, but have to show the equation in each step.

$$b_2 = z_{2,1} + z_{2,2} + z_{2,3} = \hat{\alpha}_{2,1}v_1 + \hat{\alpha}_{2,2}v_2 + \hat{\alpha}_{2,3}v_3 \quad (4)$$

3 [10 points]. In practice, it is common to have multiple self-attention operations happening in one layer. Each individual self-attention component is referred as a head, and thus the entire block is typically called Multi-head Self-Attention. What's the benefit of having multiple heads? (Hint: why do we want multiple kernel filters in CNN?)

Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions.

Problem 2 [30 points] K-means Clustering

Consider the following set of one-dimensional points: $\{5, 7, 16, 18, 24, 26, 34, 38\}$. Suppose we apply k-means clustering to obtain three clusters: A, B, and C. The overall sum-of-squared errors (SSE) of clustering is calculated by

$$\text{SSE} = \sum_{x_i \in A} (x_i - \mu_1)^2 + \sum_{x_i \in B} (x_i - \mu_2)^2 + \sum_{x_i \in C} (x_i - \mu_3)^2 \quad (5)$$

1 [5 points]. If the three initial centroids are located at $(15, 25, 31)$, respectively, show the clustering results after assigning each point to their closest centroid.

$\{5, 7, 16, 18\}, \{24, 26\}, \{34, 38\}$

2 [5 points]. Based on your answer in part (1), recompute the new locations of the centroids for A, B, and C. Compute also their overall SSE.

New centroids are 11.5, 25, 36. SSE is 135.

3 [5 points]. What are the locations of the cluster centroids when the algorithm converges? Compute also their overall SSE.

Same as part (b).

4 [7.5 points]. For the dataset given in part (1), is it possible to obtain empty clusters? If possible, what are the values of the initial centroids? If not, state why.

Yes. For example, if the centroids are located at 50, and 51. There won't be any points assigned to cluster C.

5 [7.5 points]. Suppose we add 199 equally spaced data points between 5 and 7 (i.e., 5.01, 5.02, ..., 6.99) to the previous data set. How do the clusters look like after applying k-means with $k = 3$? Suppose we add 199 equally spaced data points between 5 and 7 (i.e., 5.01, 5.02, ..., 6.99) to the previous data set.

How do the clusters look like after applying kmeans with $k = 3$?

$\{5, \dots, 7\}, \{16, 18, 24, 26\}, \{34, 38\}$

Problem 3 [20 points] Hierarchical Clustering

Use the distance matrix shown in the Figure 2 below to perform single and complete link hierarchical clustering. Show your results by drawing a dendrogram. The dendrogram should clearly show the order in which the points are merged and the y-axis show the distance between pairs of clusters being merged at each iteration. (*Hint: start with each point as a cluster, search the distance matrix from the smallest to the largest, identify the key linkage and make merge.*)

	p1	p2	p3	p4	p5
p1	0	0.3830	0.1474	0.3466	0.1616
p2	0.3830	0	0.4592	0.1252	0.2227
p3	0.1474	0.4592	0	0.3840	0.2643
p4	0.3466	0.1252	0.3840	0	0.2102
p5	0.1616	0.2227	0.2643	0.2102	0

Figure 2: Distance matrix.

1 [10 points]. Dendrogram by single link

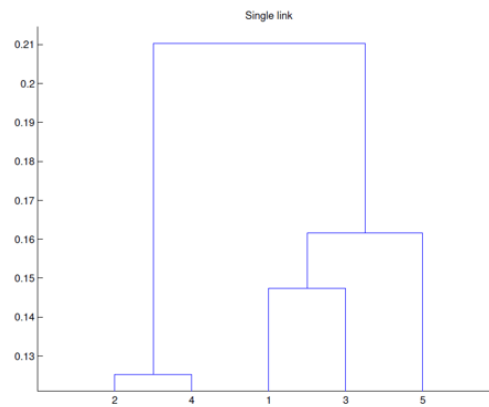


Figure 8.1. Single link.

2 [10 points]. Dendrogram by complete link

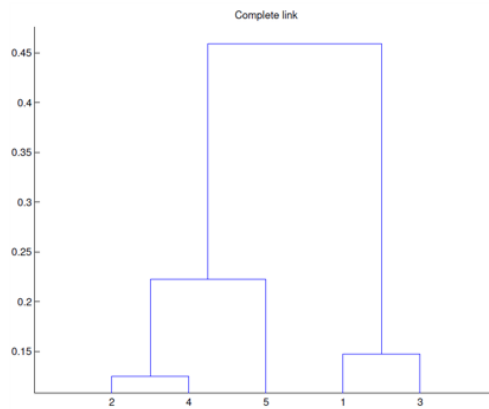


Figure 8.2. Complete link.

Problem 4 [20 points] Bayes Rule

Smith owns a retail store for selling phones. The phones are manufactured at three different factories: A, B , and C . Factory A , B and C produces 20%, 30%, and 50% of the phone being sold at Smith's store. The probabilities of the defective phones from factory A , B and C are 2%, 1%, and 0.05%, respectively. The total number of phones being sold at Smith's store is 10000. One day, a customer walks up to Smith's store, and ask for a refund for a defective phone.

1 [5 points]. What is the probability of a phone being defective $P(D)$?

The probability of a phone being defective is

$$\begin{aligned} P(D) &= P(D|A)P(A) + P(D|B)P(B) + P(D|C)P(C) \\ &= 0.02 \times 0.2 + 0.01 \times 0.3 + 0.0005 \times 0.5 \\ &= 0.00725 \end{aligned} \tag{6}$$

2 [5 points]. What is the probability that this defective phone is manufactured at factory A ? Hint: $P(A|D)$

$$\begin{aligned} P(A|D) &= \frac{P(A, D)}{P(D)} \\ &= \frac{P(D|A)P(A)}{P(D)} \\ &= \frac{0.02 \times 0.2}{0.00725} \\ &= 0.55 \end{aligned} \tag{7}$$

3 [5 points]. What is the probability that this defective phone is manufactured at factory B ?

$$\begin{aligned} P(B|D) &= \frac{P(B, D)}{P(D)} \\ &= \frac{P(D|B)P(B)}{P(D)} \\ &= \frac{0.01 \times 0.3}{0.00725} \\ &= 0.41 \end{aligned} \tag{8}$$

4 [5 points]. What is the probability that this defective phone is manufactured at factory C ?

$$\begin{aligned} P(C|D) &= \frac{P(C, D)}{P(D)} \\ &= \frac{P(D|C)P(C)}{P(D)} \\ &= \frac{0.0005 \times 0.5}{0.00725} \\ &= 0.034 \end{aligned} \tag{9}$$